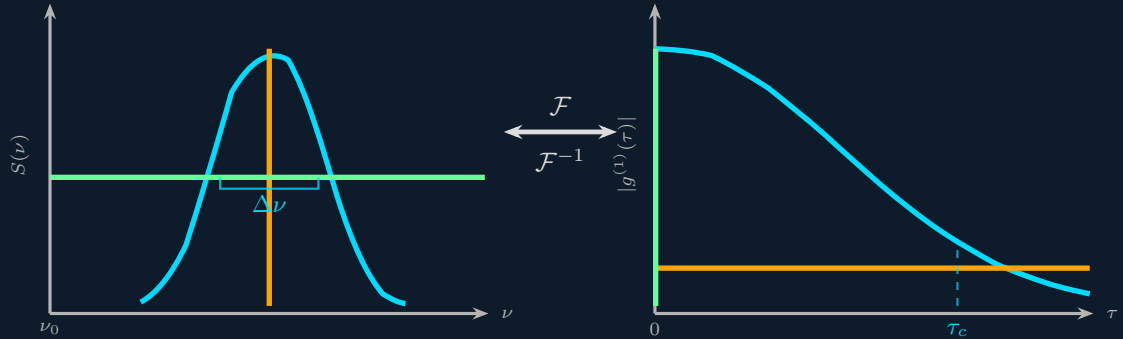


Source First-Order Coherence — Wiener-Khinchin Theorem — $g^{(1)}(\tau) \Leftrightarrow S(\nu)$



$$E(t) = \int_{-\infty}^{\infty} \tilde{E}(\nu) e^{-i2\pi\nu t} d\nu$$

$$\Gamma(\tau) = \langle E^*(t) E(t + \tau) \rangle = \left\langle \int \int \tilde{E}^*(\nu) \tilde{E}(\nu') e^{i2\pi(\nu - \nu')t} e^{-i2\pi\nu'\tau} d\nu d\nu' \right\rangle$$

$$\langle e^{i2\pi(\nu - \nu')t} \rangle = \delta(\nu - \nu') \implies \Gamma(\tau) = \int |\tilde{E}(\nu)|^2 e^{-i2\pi\nu\tau} d\nu$$

$$g^{(1)}(\tau) = \frac{\Gamma(\tau)}{\Gamma(0)} = \frac{\int S(\nu) e^{-i2\pi\nu\tau} d\nu}{\int S(\nu) d\nu}$$